

```
public class Book
{
    public String title;
    public double price;
    public String ISBN;
    public String genre; // fiction, fantasy, mystery, thriller, horror
    public String author;

    public Book()
    {
        title = "The Silent Library";
        price = 399.50;
        ISBN = "978-1-4028-9462-6";
        genre = "fiction";
        author = "Ava Reynolds";
    }

    public Book(String t, double p, String i, String g, String a) throws
    InvalidPriceException, InvalidTitleException
    {
        if(t == null)
            throw new InvalidTitleException("title cannot be empty or null.");

        if(p < 0)
            throw new InvalidPriceException("price cannot be negative.");

        title = t;
        price = p;
        ISBN = i;
        genre = g;
        author = a;
    }

    public Book(Book b)
    {
        title = b.title;
        price = b.price;
        ISBN = b.ISBN;
        genre = b.genre;
        author = b.author;
    }
}
```

```

public class InvalidPriceException extends Exception
{
    public InvalidPriceException(String msg)
    {
        super(msg);
    }
}

public class InvalidTitleException extends Exception
{
    public InvalidTitleException(String msg)
    {
        super(msg);
    }
}

```

```

import java.util.ArrayList;
public class ArrayListOfBooks
{
    public static void main(String[] args)
    {
        Book b1 = new Book();

        Book b4 = new Book(b1);
        b4.title = "The Last Clue";
        b4.price = 299.99;

        ArrayList<Book> books = new ArrayList<Book>();
        books.add(b1);
        books.add(b4);

        try
        {
            Book b2 = new Book("Realms of Neverland", 549.00, "978-0-545-01022-1", "fantasy", "Liam Carter");
            Book b3 = new Book(b2);
            b3.title = "Whispers in the Dark";
            b3.price = 375.00;
            b3.author = "Enid Blyton";

            books.add(b2);
            books.add(b3);

            Book b5 = new Book(null, -450.75, "978-1-86197-876-9", "thriller", "Ethan Brooks");

```

```

        books.add(b5);
    }

    catch(InvalidPriceException ipe)
    {
        System.out.println(ipe.getMessage());
    }

    catch(InvalidTitleException ite)
    {
        System.out.println(ite.getMessage());
    }

    books.forEach(e ->
    {
        System.out.println("\nTitle: " + e.title);
        System.out.println("Price: " + e.price);
        System.out.println("ISBN: " + e.ISBN);
        System.out.println("Genre: " + e.genre);
        System.out.println("Author: " + e.author);
    });

    // average price calculation
    double sum = 0;
    for(Book b : books)
        sum += b.price;
    double avg = sum / books.size();
    System.out.println("Average Price: " + avg);

    // printing fiction books
    System.out.println("\nFiction Books:");
    books.forEach(b ->
    {
        if(b.genre.equalsIgnoreCase("fiction"))
            System.out.println(b.title + " - " + b.author);
    });
}
}

```

```
PS C:\Users\hp\Downloads\Programming in Java> cd "c:\Users\hp\Downloads\Programming in Java\" ; if ($?) {  
javac ArrayListOfBooks.java } ; if ($?) { java ArrayListOfBooks }  
title cannot be empty or null.
```

```
Title: The Silent Library  
Price: 399.5  
ISBN: 978-1-4028-9462-6  
Genre: fiction  
Author: Ava Reynolds
```

```
Title: The Last Clue  
Price: 299.99  
ISBN: 978-1-4028-9462-6  
Genre: fiction  
Author: Ava Reynolds
```

```
Title: Realms of Neverland  
Price: 549.0  
ISBN: 978-0-545-01022-1  
Genre: fantasy  
Author: Liam Carter
```

```
Title: Whispers in the Dark  
Price: 375.0  
ISBN: 978-0-545-01022-1  
Genre: fantasy  
Author: Enid Blyton  
Average Price: 405.8725
```

Code

Code

Fiction Books:

The Silent Library - Ava Reynolds

The Last Clue - Ava Reynolds

PS C:\Users\hp\Downloads\Programming in Java> █